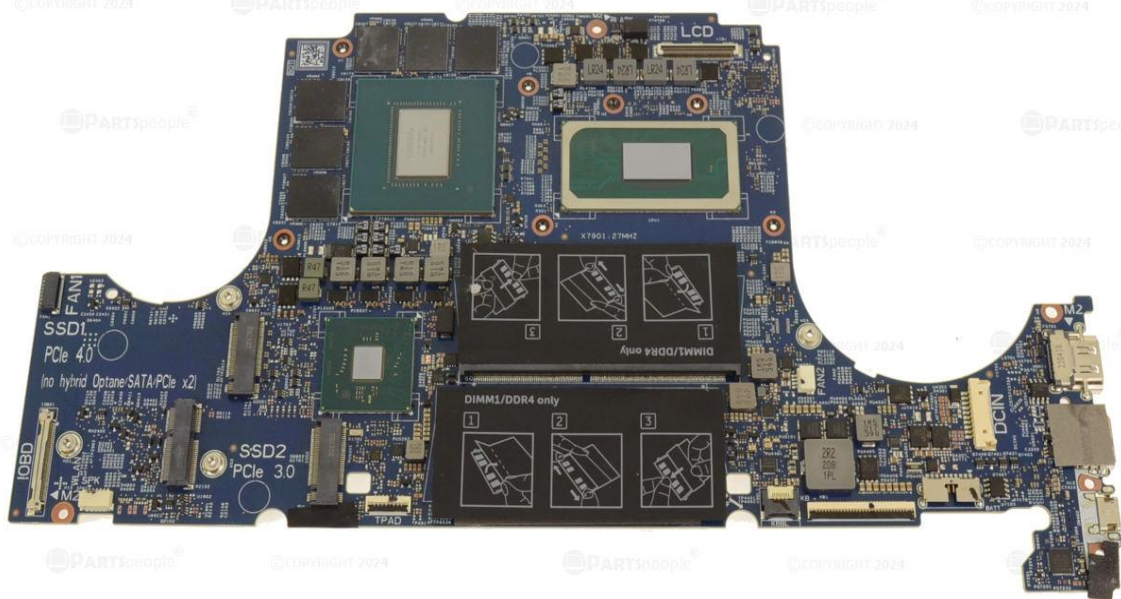


1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	1.5V (1.35V for DDR3L)	1.2V	1.1V
Power Consumption	Higher	Lower than DDR3	Lowest among the three
Bandwidth	Lower	Higher than DDR3	Highest
Maximum Capacity per Module	Up to 16 GB	Up to 64 GB	Up to 128 GB or more
Performance	Suitable for older systems	Good for modern PCs and gaming	Best for high-performance computing and latest systems
Typical Use Cases	Older desktops and laptops, basic computing	Gaming PCs, office computers, workstations	High-end gaming, AI, content creation, and advanced workstations
Compatibility	DDR3 motherboards only	DDR4 motherboards only	DDR5 motherboards only

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.



3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

HDD vs SSD vs NVMe Storage Comparison

Feature	HDD (Hard Disk Drive)	SSD (Solid-State Drive)	NVMe SSD
Speed	Slow (50–100 MB/s)	Fast (100–500+ MB/s)	Very Fast (up to several GB/s)
Technology	Uses spinning magnetic disks	Uses flash memory chips	Uses flash memory and PCIe interface
Cost	Cheapest per GB	More expensive than HDD	Most expensive among the three
Typical Use Case	Large file storage and backups	Everyday computing and gaming	High-performance tasks and professional work

Apps / Daily Use Cases That Benefit Most

Storage Type	Example App / Use Case	Benefit
HDD	Movie and Photo Storage	Large storage capacity at low cost
SSD	Spotify, Chrome, Microsoft Office	Faster startup and smoother performance
NVMe SSD	Adobe Premiere Pro, AutoCAD, Modern Games (PUBG PC)	Extremely fast loading, file transfers, and processing

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.